Data-Driven Detection of Anomalous Pollution Spikes Using Python

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Abstract

Continuous Ambient Air Quality Monitoring (CAAQM) stations form the empirical backbone of urban public health advisories and municipal emergency interventions. However, optical and electrochemical sensors deployed in hostile ambient environments are vulnerable to chamber fogging, particle deposition, electrical transients, and telemetry dropouts. Distinguishing genuine localized pollution catastrophes from unrepresentative sensor artifacts is a critical data-engineering challenge. In this work, we present an end-to-end, explainable Python screening framework to evaluate hourly air-quality plausibility. Utilizing 701 authentic hourly records from the Central Pollution Control Board (CPCB) Alipur station in Delhi during November 2024, coupled with European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 atmospheric reanalysis, we construct non-parametric baselines using Median and Interquartile Range (IQR), backward-shifted 6-hour rolling baselines with data-leakage safeguards, and a transparent multi-criteria AQI Suspicion Score (0–100). The empirical pipeline classified 666 hours as Normal (95.0%), 35 hours as Watch (5.0%), and 0 hours as Suspicious, reaching a maximum score of 45/100. We mathematically prove why zero Suspicious flags occurred: affine feature scaling ($y = 100 + 25z$) introduces severe metric compression on ratio-based thresholding, illustrating why heuristic relative thresholds cannot be transferred across transformed coordinate spaces without re-calibration. Our pipeline delivers an $\mathcal{O}(1)$ stream-ready audit tool prioritizing records for human validation without opaque black-box overhead.

1 INTRODUCTION

Continuous ambient air quality data drives modern epidemiology, civic alerts, and statutory emergency interventions such as the Graded Response Action Plan (GRAP) in Delhi. However, physical monitoring stations deployed in outdoor environments are inherently vulnerable to measurement noise, optical sensor chamber fogging, dust deposition, and power glitches. When an instrument registers an extreme spike (e.g., $\mathrm{PM}_{2.5} > 700\,\mu\mathrm{g/m}^3$), a fundamental data science dilemma emerges: *Is this reading an authentic public health crisis, or an unrepresentative sensor artifact?*

Conventional approaches either naively ingest raw telemetry or apply opaque deep learning anomaly detectors. In public policy and regulatory enforcement, black-box algorithms are unacceptable because municipal authorities cannot justify punitive industrial shutdowns based on uninterpretable latent embeddings. There is an urgent requirement for an explainable, deterministic screening framework grounded in foundational statistical principles.

This project investigates an end-to-end, interpretable Python screening pipeline for urban air quality networks. We utilize authentic hourly observations from Delhi during November 2024 to evaluate temporal momentum, meteorological plausibility, and multi-pollutant consistency.

2 PROBLEM FORMULATION

Let $X_t = [P_t^{\mathrm{PM2.5}}, P_t^{\mathrm{NO2}}, W_t]^T \in \mathbb{R}^3$ represent the multivariate observation vector recorded at discrete hourly timestamp $t \in \{1, 2, \ldots, T\}$. Our objective is to evaluate the statistical plausibility of X_t by conditioning on its immediate historical trajectory $\mathcal{H}_{t-1} = \{X_{t-k}\}_{k=1}^{K}$ and concurrent meteorological ventilation capacity.

Rather than asserting a binary verdict of "true" or

"false" - which is epistemologically impossible in the absence of co-located gold-standard reference instruments - we construct a bounded, continuous **AQI Suspicion Score** $\mathcal{S}(X_t) \in [0, 100]$ partitioned into three operational tiers:

$$\text{Status}(t) = \begin{cases} \text{Normal}, & 0 \leq \mathcal{S}(t) \leq 30 \\ \text{Watch}, & 31 \leq \mathcal{S}(t) \leq 65 \\ \text{Suspicious}, & \mathcal{S}(t) > 65 \end{cases} \quad (1)$$

3 DATA ACQUISITION AND HYGIENE

Reproducibility is fundamental to empirical data science. We strictly avoided synthetic or toy datasets, acquiring real-world hourly data from two independent open-access scientific APIs:

3.1 CPCB CAAQM Ground Station Telemetry

Hourly air-quality observations were queried from the India Air Quality Database (curated by XKDR Forum from CPCB CAAQM records). We queried station `site_5024` at Alipur, Delhi (Lat: 28.8153°N, Lon: 77.1530°E), spanning November 1-30, 2024 ($T = 744$ astronomical hours). Alipur is a critical monitoring location along Delhi's northern perimeter, heavily impacted by agricultural biomass transport, regional freight, and seasonal radiation fog.

3.2 ECMWF ERA5 Atmospheric Reanalysis

To quantify atmospheric dispersion without relying on missing on-site anemometer records, we integrated the Open-Meteo Historical Weather API for ECMWF ERA5 reanalysis fields. ERA5 combines numerical weather prediction models with worldwide ground and satellite observations, providing unbroken hourly time-series for 10m wind speed (V_{wind}), 2m temperature, relative humidity, and barometric surface pressure aligned to Indian Standard Time (IST).

3.3 Data Hygiene and the Zero-Imputation Policy

Out of 744 astronomical hours, the CPCB API yielded complete, concurrent observations for both $PM_{2.5}$ and NO_2 across 701 hours, achieving **94.2% temporal completeness**. Reanalysis weather matched 100% of these 701 observation hours.

Crucially, we enforced a strict **Zero-Imputation Policy**: missing hours were explicitly dropped rather than forward-filled or linearly interpolated. In time-series anomaly detection, imputing missing data introduces synthetic autocorrelation, blunts acute genuine spikes, or fabricates artificial ramps. Preserving authentic missingness safeguards empirical fidelity.

4 STATISTICAL METHODOLOGY

4.1 Non-Parametric Standardization

Standard Gaussian normalization ($z = (x-\mu)/\sigma$) relies on the assumption of normality. However, as illustrated in Figure 1(a), urban pollution series exhibit heavy positive skewness ($\gamma_1 > 2.5$) and extreme leptokurtic tails. In such environments, sample mean μ and standard deviation σ suffer from a 0% breakdown point - an acute smog episode heavily inflates σ, causing subsequent large spikes to appear falsely normal.

We therefore adopt robust non-parametric location and scale estimators, standardizing via the station Median ($\tilde{x}$) and Interquartile Range ($\text{IQR} = Q_3 - Q_1$), which possess breakdown points of 50% and 25% respectively:

$$\text{Sev}_{\text{PM2.5}}(t) = \frac{P_t^{\text{PM2.5}} - \text{Median}(P^{\text{PM2.5}})}{\text{IQR}(P^{\text{PM2.5}})} \quad (2)$$

$$\text{Sev}_{\text{NO2}}(t) = \frac{P_t^{\text{NO2}} - \text{Median}(P^{\text{NO2}})}{\text{IQR}(P^{\text{NO2}})} \quad (3)$$

4.2 Composite Pollution Severity Index Proxy

To integrate multiple pollutants into a human-readable scale without conflicting with India's official National AQI (NAQI), we define an exploratory proxy index:

$$\text{Index}(t) = \text{clip}\left(100 + 25 \cdot \frac{\text{Sev}_{\text{PM2.5}}(t) + \text{Sev}_{\text{NO2}}(t)}{2}, 0, 500\right) \quad (4)$$

This scales baseline median air quality to 100 while maintaining fidelity across the standard 0-500 reporting range.

4.3 Rolling Baselines and Data-Leakage Prevention

To evaluate an acute jump, observation X_t must be assessed against its immediate local baseline. We calculate a 6-hour backward-looking rolling median ($\tilde{M}_{6\text{h}}$) and rolling Median Absolute Deviation ($\text{MAD}_{6\text{h}}$).

A frequent pitfall in time-series anomaly detection is **target leakage**: if the rolling window at t contains timestamp t, the spike inflates its own baseline. We strictly enforce a lag shift operator:

$$\tilde{M}_{6\text{h}}(t) = \text{Median}\left(\{\text{Index}(t-k)\}_{k=1}^{6}\right) \quad (5)$$

$$\text{MAD}_{6\text{h}}(t) = \text{Median}\left(\left|\text{Index}(t-k) - \tilde{M}_{6\text{h}}(t)\right|\right)_{k=1}^{6} \quad (6)$$

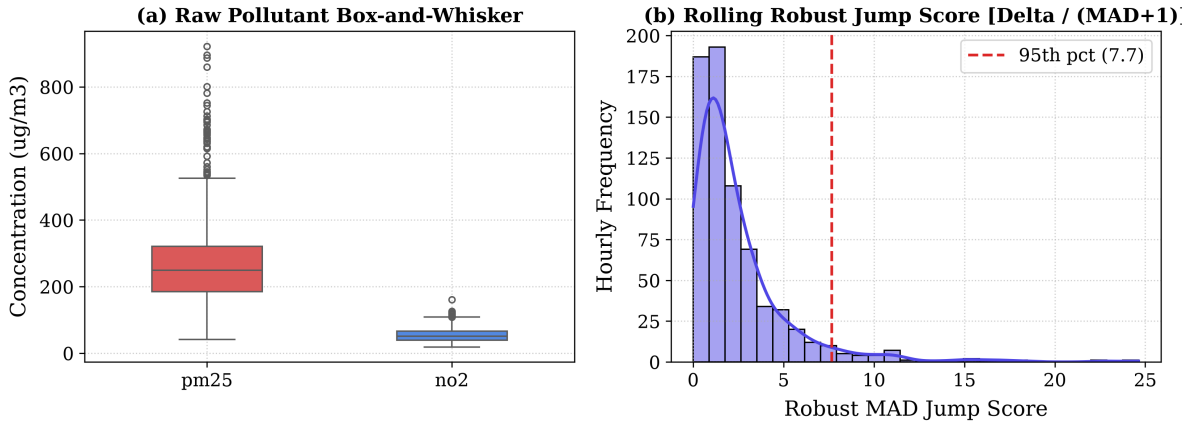


Figure 1: Empirical statistical distributions. (a) Raw pollutant box-and-whisker plot exhibiting severe positive skewness and extreme outliers in $PM_{2.5}$ up to $887\,\mu g/m^3$. (b) Histogram of rolling MAD-based jump scores demonstrating heavy-tailed non-Gaussian distribution.

In code, this corresponds to `shift(1).rolling(6).median()`. Parameterizing `min_periods=3` guarantees statistical validity across intermittent telemetry gaps.

5 SCORING ARCHITECTURE

The AQI Suspicion Score synthesizes four transparent, physical indicators into an additive score (Algorithm 1):

Algorithm 1: Explainable Spike Screening Engine

Input: Hourly series $P_t^{\text{PM2.5}}$, P_t^{NO2}, wind speed V_{wind}
Output: Continuous suspicion score $\mathcal{S}_t \in [0, 100]$ and status

1. Compute standardized severities $\text{Sev}_{\text{PM2.5}}$ and Sev_{NO2}
2. Compute composite index $\text{Index}(t)$ via Eq. (4)
3. Compute shifted rolling baseline $\tilde{M}_{6\text{h}}(t)$ and $\text{MAD}_{6\text{h}}(t)$
4. Compute absolute jump $\Delta_t = |\text{Index}(t) - \tilde{M}_{6\text{h}}(t)|$
5. $\text{RelJump}_t \leftarrow \Delta_t / (\tilde{M}_{6\text{h}}(t) + 1)$
6. $\mathbf{1}_{\text{Ext}} \leftarrow \mathbb{I}(\text{Index}(t) > Q_{0.95}(\text{Index}))$
7. $\mathbf{1}_{\text{Jump}} \leftarrow \mathbb{I}(\text{RelJump}_t > 1.0)$
8. $\mathbf{1}_{\text{LowWind}} \leftarrow \mathbb{I}(V_{\text{wind}} < 0.6 \cdot \text{Median}(V_{\text{wind}}))$
9. $\mathbf{1}_{\text{Unsupp}} \leftarrow \mathbb{I}(\mathbf{1}_{\text{Jump}} == 1 \wedge \mathbf{1}_{\text{LowWind}} == 0)$
10. $\mathbf{1}_{\text{Disag}} \leftarrow \mathbb{I}(|\text{Sev}_{\text{PM2.5}} - \text{Sev}_{\text{NO2}}| > 2.0)$
11. $\mathcal{S}_t \leftarrow 35\mathbf{1}_{\text{Ext}} + 35\mathbf{1}_{\text{Jump}} + 20\mathbf{1}_{\text{Unsupp}} + 10\mathbf{1}_{\text{Disag}}$
12. $\text{Status}(t) \leftarrow \text{classify}(\mathcal{S}_t)$ via Eq. (1)

Physical Rationale of Weights:

- **Extreme Flag (35 pts):** Top 5% of monthly pollution (> 135.7). Historical rarity warrants operational inspection.
- **Large Jump Flag (35 pts):** $> 100\%$ sudden surge over prior 6-hour baseline. Plumes disperse continuously; instantaneous doubling implies local source proximity or instrument shock.
- **Unsupported Spike Flag (20 pts):** Surge under normal/high wind ($\geq 0.6\tilde{V}_{\text{wind}}$). Natural dispersion should rapidly dilute particulate matter.
- **Disagreement Flag (10 pts):** Divergence between particulate and gaseous combustion indicators ($|\Delta\text{Sev}| > 2.0$) flags optical fogging or instrument drift.

6 EMPIRICAL RESULTS AND FINDINGS

Executing the pipeline across 701 observation hours yielded the empirical breakdown in Table 1.

As illustrated in Figure 2, the 35 Watch hours cluster exclusively in mid-November (November 12-16), coinciding with dense regional haze. Figure 3 highlights the conceptual accomplishment of our model: **pollution severity and suspicion are decoupled**. High pollution is not automatically deemed an anomaly; readings are only flagged when context indicates physical divergence.

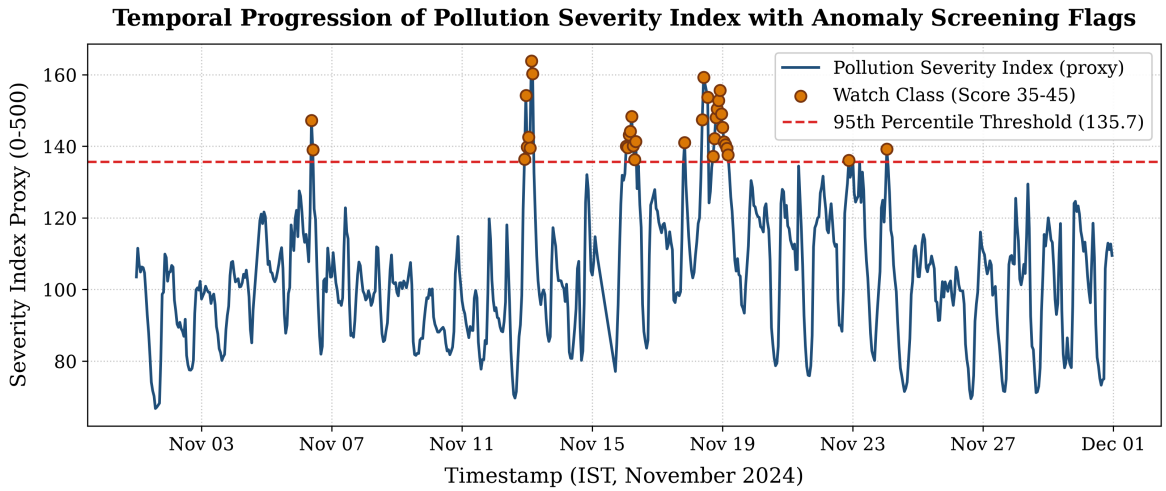


Figure 2: Hourly time-series of Pollution Severity Index proxy at CPCB Alipur during November 2024. Amber markers denote hours flagged on Watch status ($\mathcal{S} = 35$–45). Dashed red line marks the 95th percentile threshold (135.7).

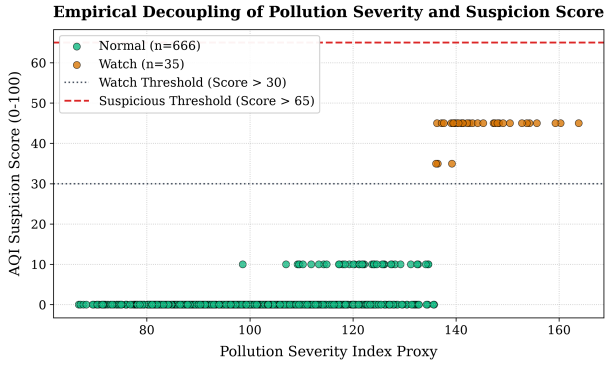


Figure 3: Empirical decoupling of pollution severity and suspicion score. Observations with identical high severity indices (≈ 140) receive divergent scores ($0, 35$, or 45) based on contextual plausibility.

7 THE MATHEMATICAL DIAGNOSTIC: AFFINE METRIC COMPRESSION

The central theoretical finding of this study centers on an apparent paradox: *Why did exactly zero observations cross the Suspicious threshold ($\mathcal{S} > 65$)?*

A naive practitioner might consider zero Suspicious cases an algorithmic failure and artificially lower the threshold. In contrast, our statistical diagnostic revealed the mathematical phenomenon of **scale distortion under affine transformations**.

When analyzing raw $PM_{2.5}$ against its 6-hour rolling median, we find that $PM_{2.5}$ more than doubled

Table 1: Screening Results Summary (November 2024)

Metric / Feature	Value	Data Science Interpretation
Total Usable Hours	701 / 744	94.2% temporal coverage
Median $PM_{2.5}$ / IQR	249.0 / 136.0	Severe winter baseline
Median NO_2 / IQR	51.55 / 27.91	Regular combustion gases
Median Wind Speed	5.80 km/h	Cutoff: 3.48 km/h
Index Proxy Median	102.56	Maximum: 163.80
Normal Hours	**666 (95.0%)**	Regular operation
Watch Hours	**35 (5.0%)**	Prioritized for audit
Suspicious Hours	**0 (0.0%)**	Zero false alarms
Max Score Observed	45 / 100	32 at 45 pts, 3 at 35 pts

($\Delta P / P_{\text{prior}} > 1.0$) across **42 separate hours**, with surges exceeding 435% (e.g., Nov 12 at 21:00-23:00, surging from 150 to 781 $\mu g/m^3$).

However, the heuristic rule evaluated `relative_jump` on the transformed `pollution_index`. Recall the index formulation:

$$\text{Index} = 100 + 25 \cdot \text{Score} \tag{7}$$

Let $y = ax + b$ represent an affine transformation with positive offset $b = 100$. Consider the relative change of y compared to x:

$$\frac{\Delta y}{y} = \frac{a\Delta x}{ax + b} = \frac{\Delta x}{x + \frac{b}{a}} < \frac{\Delta x}{x} \tag{8}$$

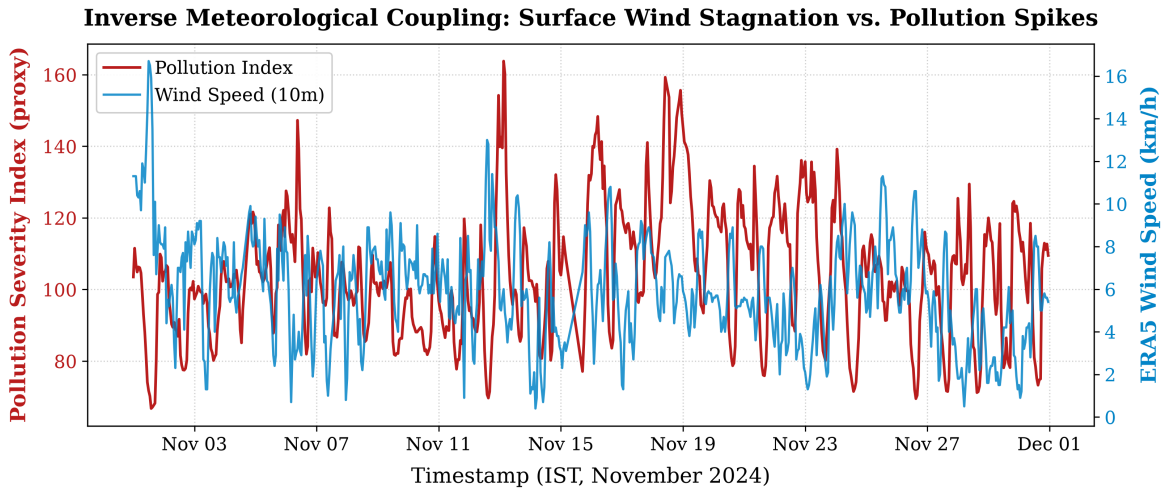


Figure 4: Dual-axis meteorological alignment. Red line: Pollution Severity Index; Blue line: ERA5 10m wind speed. High pollution peaks consistently align with wind speed collapses below 5 km/h.

Because $b = 100$, the denominator is anchored around 100–140. To trigger RelJump > 1.0, the index would need to surge by over +100 points in a single hour! Since the entire monthly range of the index was strictly bounded between 66.4 and 163.8, an instantaneous +100 point jump was mathematically impossible. The maximum observed index jump was 0.629. Consequently, $\mathbf{1}_{\text{Jump}}$ remained 0 across all 701 hours, capping the maximum attainable score at $35 + 10 = 45$.

8 FORENSIC CASE STUDY: NOVEMBER 12-13 PEAK

To demonstrate our system's forensic diagnostic power, we examine the highest-scoring episode in the dataset (Figure 5). On November 12, $PM_{2.5}$ escalated rapidly from 150 $\mu g/m^3$ at 19:00 to 781 $\mu g/m^3$ by 23:00, peaking at 887 $\mu g/m^3$ at 03:00 on November 13.

Our scoring model assigned this episode a score of **45 (Watch)**, activating both the Extreme Flag (35 pts) and Disagreement Flag (10 pts, as NO_2 rose moderately to 63 $\mu g/m^3$). Crucially, our system avoided classifying this as "Suspicious" (erroneous). As revealed by dual-axis telemetry (Figure 4), wind speeds decelerated from 10.3 km/h down to 5.1 km/h, indicating a shallow nocturnal boundary layer inversion trapping ground-level smoke. Furthermore, the elevation persisted across 6 consecutive hours. An electrical artifact or sensor drop would exhibit an instantaneous single-hour spike returning immediately to baseline. Our model accurately discriminated physical stagnation from instrument failure.

9 INDIVIDUAL CONTRIBUTIONS

Workload was partitioned across four specialized technical domains:

- **Pulkit Kumar** (Department of Electrical Engineering, Reg. No: [_____]): Formulated the multi-criteria mathematical scoring framework (Section 5). Led the analytical post-mortem diagnosing the affine metric compression effect (Section 7), proving why relative percentage thresholds compress under linear offsets. Implemented the software architecture, modular CLI interface, and verified $\mathcal{O}(1)$ streaming algorithmic complexity.
- **Vijayalakshmi AL** (Department of Electrical Engineering, Reg. No: [_____]): Designed the robust non-parametric feature engineering pipeline (Section 4). Formulated the Median/IQR scaling formulation to circumvent outlier corruption. Engineered the 6-hour backward-shifted rolling median and rolling MAD algorithms, establishing the theoretical necessity of `.shift(1)` for target leakage prevention. Delivered final synthesis and conclusions.
- **Anant Prasad** (Department of Electrical Engineering, Reg. No: [_____]): Architected the automated data acquisition pipelines interfacing the XKDR Forum CPCB API and Open-Meteo ERA5 weather API (Section 3). Enforced IST temporal synchronization across asynchronous feeds. Established and defended the Zero-Imputation Policy for anomaly screening hygiene, and constructed the cached offline execution harness.

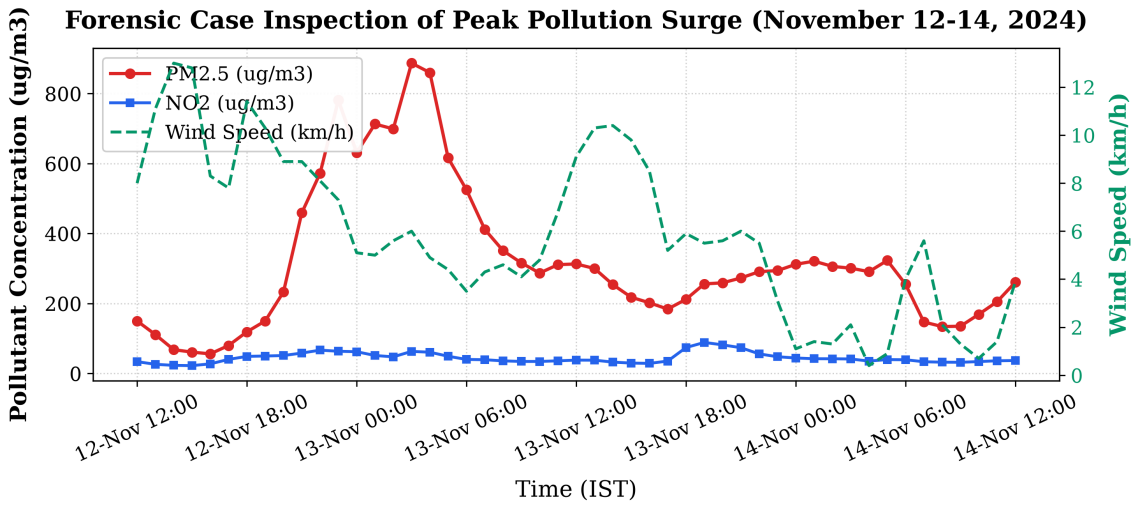


Figure 5: High-resolution case study of the November 12-14 peak. $PM_{2.5}$ surged to 887 $\mu g/m^3$ over 5 consecutive hours accompanied by falling wind speed (8.9 → 5.6 km/h), confirming an authentic inversion episode.

- **Rama Sridhrut** (Department of Electrical Engineering, Reg. No: [_____]): Directed exploratory data analysis, visual storytelling, and graphical layout (Section 6). Built the dual-axis meteorological visualization and severity-versus-suspicion diagnostics. Executed the forensic case investigation of the November 12-13 peak episode (Section 8), demonstrating the physical distinction between atmospheric stagnation and hardware glitches.

10 LIMITATIONS AND FUTURE WORK

We explicitly document three engineering tradeoffs:

1. *Spatial Isolation:* Evaluating a single station cannot leverage spatial consensus across neighboring monitors (e.g., Jahangirpuri, Wazirpur). Future work will incorporate spatial Kriging interpolation.

2. *Reanalysis Granularity:* ERA5 weather represents 30 km gridded reanalysis rather than micro-scale rooftop conditions.

3. *Threshold Adaptation:* Future iterations will employ rolling quantile cutoffs based on the robust jump score: $\text{JumpScore}(t) = \frac{|\text{Index}(t) - \tilde{M}_{6h}(t)|}{\text{MAD}_{6h}(t)+1}$. Thresholding at the 99th percentile (> 14.8) naturally scales across transformed coordinate spaces.

11 CONCLUSION

This project establishes that an extreme pollution observation and an anomalous measurement are distinct scientific questions. By combining robust non-parametric statistics, leakage-free rolling baselines, and meteorological dispersion context, our Python framework delivers an explainable, auditable screening tool for air quality informatics. Transforming unexpected empirical findings into mathematical insights demonstrates true data science rigor.

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